

ECON 3201

Econometrics for Economics and Finance

Assignment 1
Due Thursday October 8, 2026

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Instructions

Answer all questions. Please complete your assignment using Quarto. Submit both your Quarto file and pdf output file.

Theory Questions

Q1

Create a personal Quarto website hosted on GitHub pages. For reference, please see the slides for **website** on Nexus. Your website should include the following:

- i. An “Index” or “Home” page. This page should include the following:
 - Your name
 - Your photo (optional)
 - Your credentials
 - An “About Me” section that provides a brief (one paragraph) introduction to yourself. [10 Marks]
- ii. A blog page. This will be the landing site for future assignments [5 Marks]
- iii. Create a blog post and upload the finished PDF for this assignment. [3 Marks]
- iv. Navigation bar with “Home” and “Blog” buttons. The navigation bar is edited in the “_quarto.yml” file. [2 Marks]

https://yingx436-web.github.io/econ_3201_assignment_1_Chen_demo/

Q2

Write your solutions using LaTeX. Show your work. You may use R to compute the answers.

Let $X = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12\}$ be a uniformly distributed variable with probability distribution function $f_X(x) = 1/12$.

(a) Check that $f_X(x)$ is a true probability distribution.

$$\begin{aligned}\sum_{x=1}^{12} f_X(x) &= 1 \\ \sum_{x=1}^{12} f_X(x) &= \sum_{x=1}^{12} \frac{1}{12} \\ &= 12 \left(\frac{1}{12} \right) \\ &= 1 \\ 1 &= 1\end{aligned}$$

(b) Compute the expected value of X .

$$\begin{aligned}E(X) &= \sum_{x=1}^{12} x f_X(x) \\ &= \frac{1 + 2 + 3 + 4 + 5 + 6 + 7 + 8 + 9 + 10 + 11 + 12}{12} \\ &= \frac{78}{12} \\ &= 6.5\end{aligned}$$

(c) Compute the variance of X .

$$\begin{aligned}\text{Var}(X) &= \sum_{x=1}^{12} (x - E(X))^2 f_X(x) \\ &= \frac{(1 - 6.5)^2 + (2 - 6.5)^2 + (3 - 6.5)^2 + \dots + (12 - 6.5)^2}{12} \\ &= \frac{143}{12} \\ &= 11.9167\end{aligned}$$

Q3

Let $X \sim N(4, 5)$. Find the probability of each event:

(a) $Pr(X \leq 6)$

$$\begin{aligned} Pr(X \leq 6) &= Pr\left(Z \leq \frac{6-4}{\sqrt{5}}\right) \\ &= \Phi(0.89) \\ &= 0.8133 \end{aligned}$$

(b) $Pr(X > 4)$

$$\begin{aligned} Pr(X > 4) &= Pr\left(Z > \frac{4-4}{\sqrt{5}}\right) \\ &= Pr(Z > 0) \\ &= 1 - \Phi(0) \\ &= 1 - 0.5 \\ &= 0.5 \end{aligned}$$

(c) $Pr(-1 \leq X < 1)$

$$\begin{aligned} Pr(-1 \leq X < 1) &= Pr\left(\frac{-1-4}{\sqrt{5}} \leq Z < \frac{1-4}{\sqrt{5}}\right) \\ &= Pr(-2.24 \leq Z < -1.34) \\ &= \Phi(-1.34) - \Phi(-2.24) \\ &= [1 - \Phi(1.34)] - [1 - \Phi(2.24)] \\ &= (1 - 0.9099) - (1 - 0.9875) \\ &= 0.0901 - 0.0125 \\ &= 0.0776 \end{aligned}$$

Q4

Let $Z \sim N(0, 1)$.

(a) Compute $Pr(Z \leq 0.83)$

$$\begin{aligned}Pr(Z \leq 0.83) &= \Phi(0.83) \\ &= 0.7967\end{aligned}$$

(b) Compute $Pr(-0.83 \leq Z \leq 0.83)$

$$\begin{aligned}Pr(-0.83 \leq Z \leq 0.83) &= \Phi(0.83) - \Phi(-0.83) \\ &= \Phi(0.83) - (1 - \Phi(0.83)) \\ &= 0.7967 - (1 - 0.7967) \\ &= 0.7967 - 0.2033 \\ &= 0.5934\end{aligned}$$

R Exercises

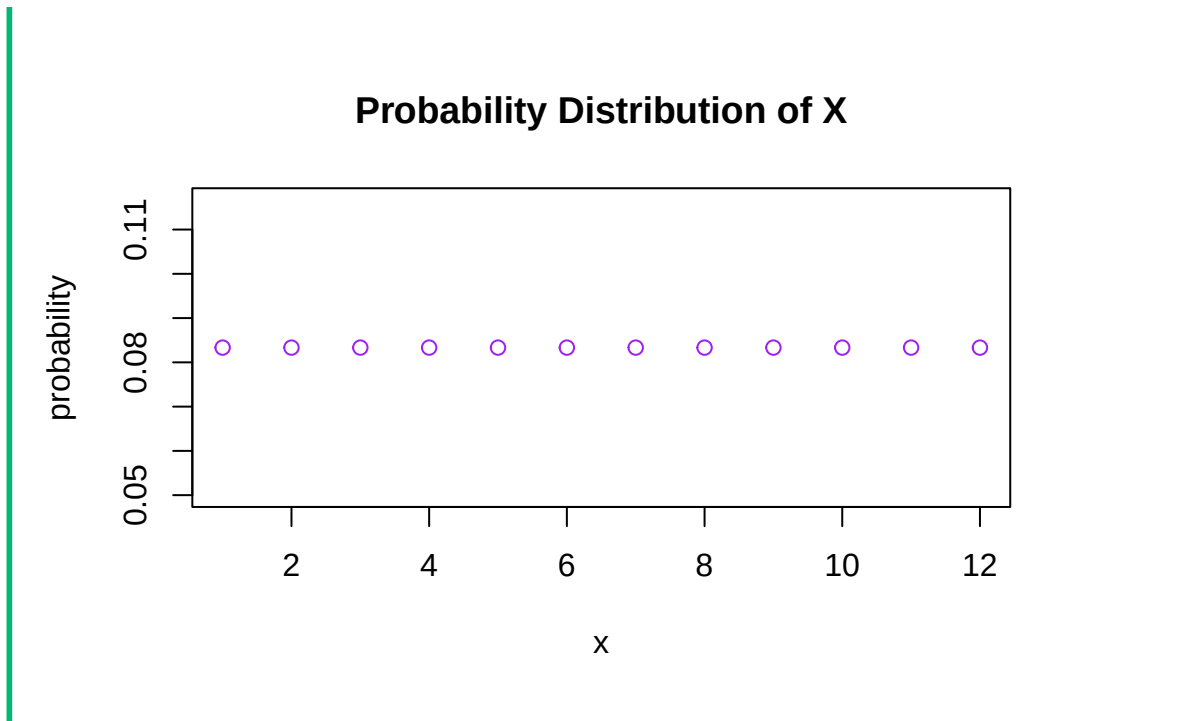
Q5

Let $X = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12\}$ be a uniformly distributed variable with probability distribution function $f_X(x) = 1/12$.

(a) Plot the probability distribution function $f_X(x)$ in R.

```
x <- 1:12
y <- rep(1/12, times = 12)

plot(x, y, main="Probability Distribution of X", xlab="x",
      ylab="probability", col="purple")
```

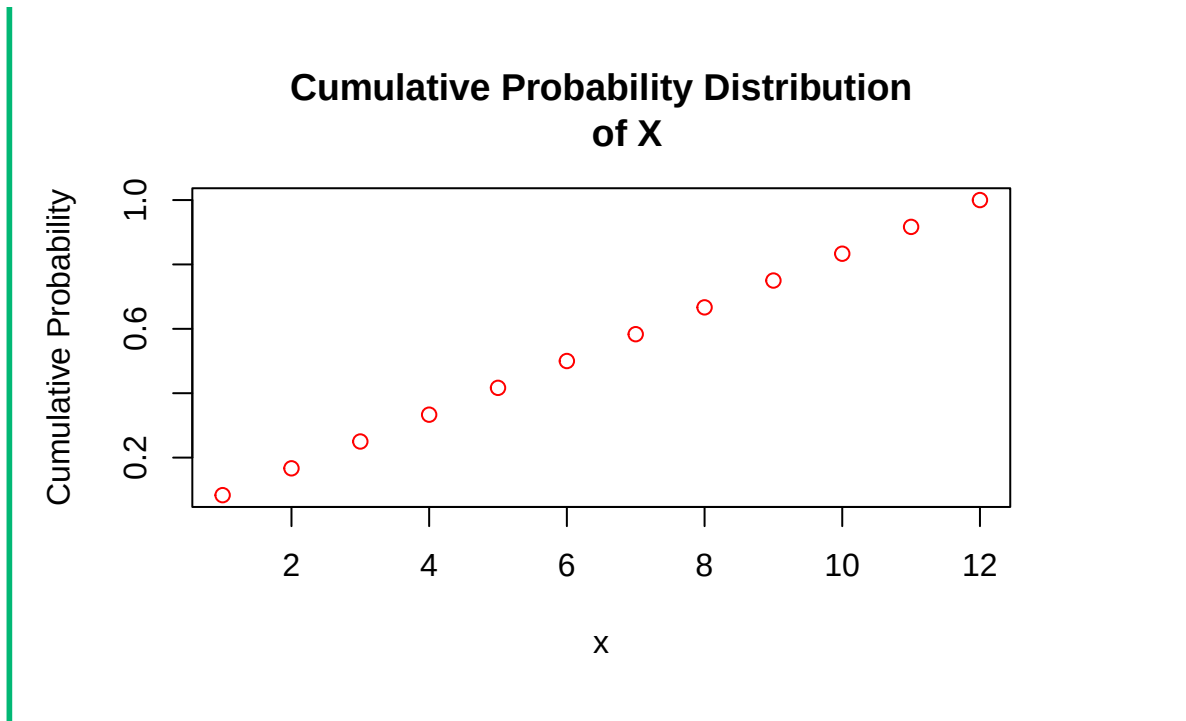


(b) Plot the cumulative probability distribution function $f_X(x)$ in R.

```
x <- 1:12
y <- rep(1/12, times = 12)

cdf <- cumsum(y)

plot(x=x, y=cdf, main="Cumulative Probability Distribution
of X", xlab="x", ylab="Cumulative Probability",
col="red")
```



(c) Compute the expected value of X in R.

```
vec <- c(1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12)
mean(vec)

[1] 6.5
```

(d) Compute the variance of X in R.

```
x <- c(1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12)
variance <- sum((x - mean(x))^2 * 1/12)
variance

[1] 11.91667
```

Q6

The standard normal distribution function is given by

$$f(x) = \frac{e^{-(x^2/2)}}{\sqrt{2\pi\sigma}}$$

- (a) Define this function in R. (Note: π is coded as `pi` and e is coded as `exp`.)

```
standard_normal <- function(x) {  
  result <- exp(-(x^2)/2) / sqrt(2 * pi * (1))  
  return(result)  
}
```

- (b) Integrate this function over the limit $[-\infty, \infty]$ using the `integrate()` command in R. Infinity in R is written as `Inf`. To evaluate the integral, use `integrate()$value`

```
integrate_normal <- integrate(standard_normal,  
  -Inf, Inf)$value  
  
integrate_normal
```

```
[1] 1
```

- (c) Compute the expected value of the standard normal distribution in R and confirm it is equal to 0.

```
expected_value_normal <- integrate(function(x)  
  x * standard_normal(x), -Inf, Inf)$value  
  
expected_value_normal
```

```
[1] 0
```

- (d) Compute the variance of the standard normal distribution in R and confirm it is equal to 1.

```
variance <- integrate(function(x)
  (x - expected_value_normal)^2 * standard_normal(x),
  -Inf, Inf)$value

variance

[1] 1
```

(e) Compute $Pr(Z \leq 0.83)$

```
pnorm(0.83, mean = 0, sd = 1)

[1] 0.7967306
```

(f) Compute $Pr(-0.83 \leq Z \leq 0.83)$

```
integrate(standard_normal, -0.83, 0.83)$value

[1] 0.5934612
```

(g) Compute $Pr(-0.83 \leq Z \leq 0.83)$ using the `pnorm()` command and compare your results to (f).

```
pnorm(0.83) - pnorm(-0.83)

[1] 0.5934612
```

Q7

Let X denote the prison sentence, in years, for people convicted of auto theft in a particular province in Canada. Suppose the PDF of X is given by:

$$f(X) = \frac{1}{9}x^2, \quad 0 < x < 3.$$

(a) Define $f(x)$ in R.

```
prison_sentence <- function(x){
  result <- x^2 /9
  return(result)
}
```

(b) Integrate $f(x)$ over the interval $[0,3]$.

```
integrate(prison_sentence, 0, 3)$value
```

```
[1] 1
```

(c) Compute the expected prison sentence.

```
expected_prison_sentence <- integrate(function(x) x *
  prison_sentence(x), 0, 3)$value
```

```
expected_prison_sentence
```

```
[1] 2.25
```

Q8

Let X follow a Student t distribution with M degrees of freedom. Then

$$X = \frac{Z}{\sqrt{W/M}} \sim t_M,$$

where Z is a standard normal variable and W is a chi-squared variable with M degrees of freedom.

(a) On the same figure plot the following:

- (i) Plot the standard normal probability distribution using `curve()`. Specify the range as $[-5,5]$ using `xlim`. Name the `ylab` “Density”. The main title should be “Densities of t Distribution”.
- (ii) Plot the Student t distribution with 5 degrees of freedom using `curve()` and `dt()`. Specify the range as $[-5,5]$ using `xlim`. Choose a different colour, i.e. `col=2`. To learn more about `dt`, type `?dt` in your console.
- (iii) Plot the Student t distribution with 15 degrees of freedom using `curve()` and `dt()`. Specify the range as $[-5,5]$ using `xlim`. Choose a different colour, i.e. `col=3`. To learn more about `dt`, type `?dt` in your console.

- (iv) Plot the Student t distribution with 25 degrees of freedom using `curve()` and `dt()`. Specify the range as `[-5,5]` using `xlim`. Choose a different colour, i.e. `col=4`. To learn more about `dt`, type `?dt` in your console.
- (v) Include the following legend in you plot:
`legend("topright", c("N(0, 1)", "M=2", "M=4", "M=25"), col = 1:4, lty = c(1, 1, 1, 1))`

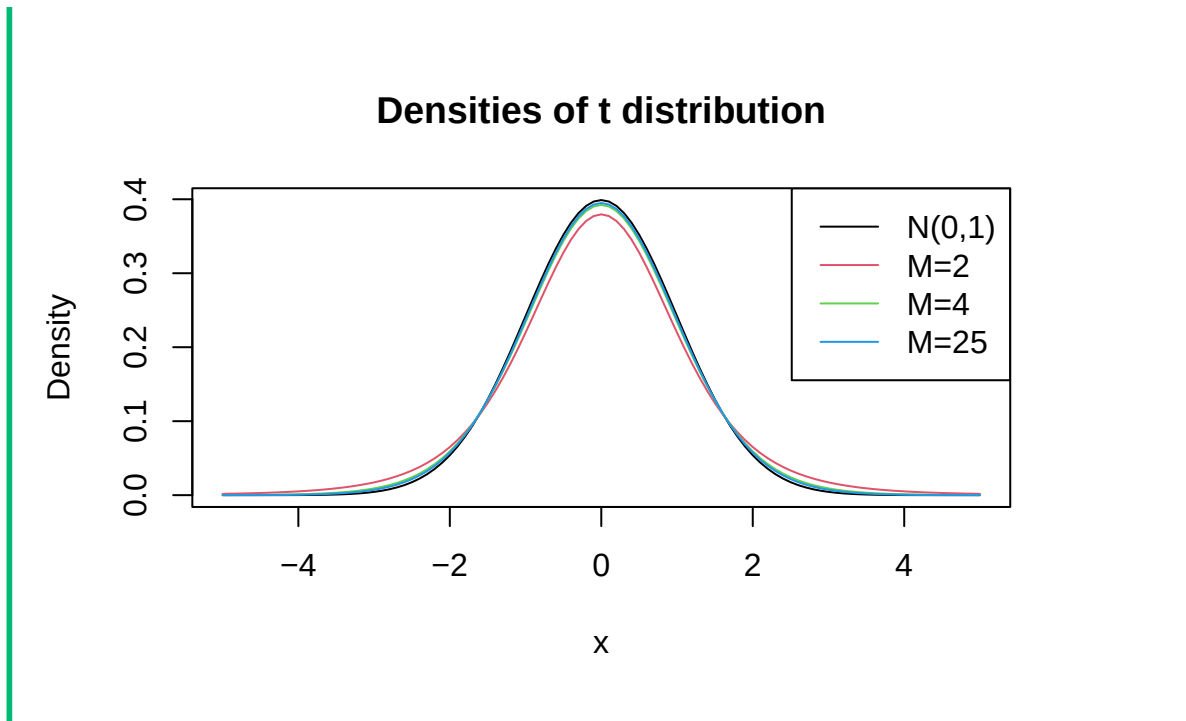
```
# Plot the PDF of a standard normal random variable
curve(dnorm(x),
      xlim = c(-5, 5),
      ylab = "Density",
      main = "Densities of t distribution")

# Student t Distribution with 5 degree of freedom
curve(dt(x, df = 5),
      xlim = c(-5,5),
      col = 2,
      add = TRUE)

# Student t Distribution with 15 degree of freedom
curve(dt(x, df = 15),
      xlim = c(-5,5),
      col = 3,
      add = TRUE)

# Student t Distribution with 25 degree of freedom
curve(dt(x, df = 25),
      xlim = c(-5,5),
      col = 4,
      add = TRUE)

legend("topright",
      legend = c("N(0,1)", "M=2", "M=4", "M=25"),
      col = 1:4,
      lty = c(1, 1, 1, 1))
```



(b) What do you observe as the degrees of freedom increase?

As M increases, the student t distribution approaches closer to the standard normal distribution.